UNIVERSITY OF INFORMATION TECHNOLOGY
FACULTY OF INFORMATION SYSTEMS

Chapter 6

DATA MODELING USING THE ENTITY-RELATIONSHIP (ER) MODEL

Content

1. Introduction
2. Entity-Relationship data model
3. Converting Entity- Relationship model to Relational data model

Introduction

- **Requirements collection and analysis**
  - Database designers interview prospective database users to understand and document data requirements
  - Result: **data requirements**
  - **Functional requirements** of the application

Introduction

- **Conceptual schema**
  - Conceptual design
  - Description of data requirements
  - Includes detailed descriptions of the entity types, relationships, and constraints
  - Transformed from high-level data model into implementation data model
- **Logical design** or **data model mapping**
  - Result is a database schema in implementation data model of DBMS
- **Physical design phase**
  - Internal storage structures, file organizations, indexes, access paths, and physical design parameters for the database files specified

Introduction

- **Conceptual schema**
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  - Description of data requirements
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Company example

- Employees, departments, and projects
- Company is organized into departments
- Department controls a number of projects
- Employee: store each employee's name, Social Security number, address, salary, sex (gender), and birth date
- Keep track of the dependents of each employee

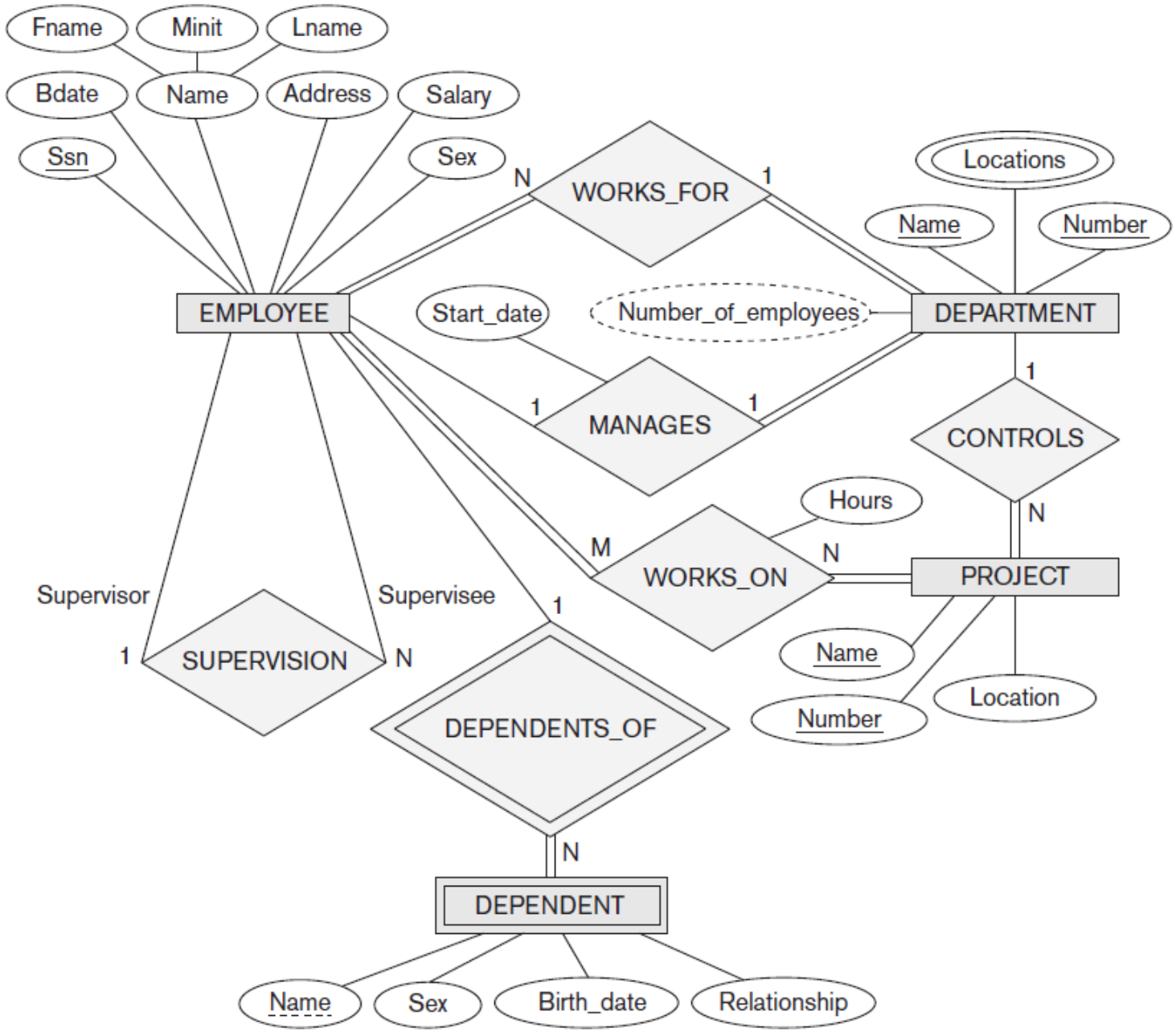

Figure 7.2
An ER schema diagram for the COMPANY database. The diagrammatic notation is introduced gradually throughout this chapter and is summarized in Figure 7.14.

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2. Entity-Relationship data model
3. Converting Entity- Relationship model to Relational data model

Entity-Relationship data model

1. Entities, Attributes, entity sets
2. Key
3. Relationship and relationship sets
4. Relationship cardinality
5. Extended Entity-relationship model

Entity-Relationship data model

- ER model describes data as:
  - Entities
  - Relationships
  - Attributes

Entities, Attributes, Entity sets

- **Entity**
  - Thing in real world with independent existence
- **Attributes**
  - Particular properties that describe entity
  - Types of attributes:
    - ✓ *Composite* versus *simple* (atomic) *attributes*
    - ✓ **Single-valued** versus **multivalued** attributes
    - ✓ **Stored** versus **derived** attributes
    - ✓ **NULL** values
    - ✓ **Complex** attributes
- **Entity type**
  - Collection (or set) of entities that have the same attributes

Entities, Attributes, Entity sets

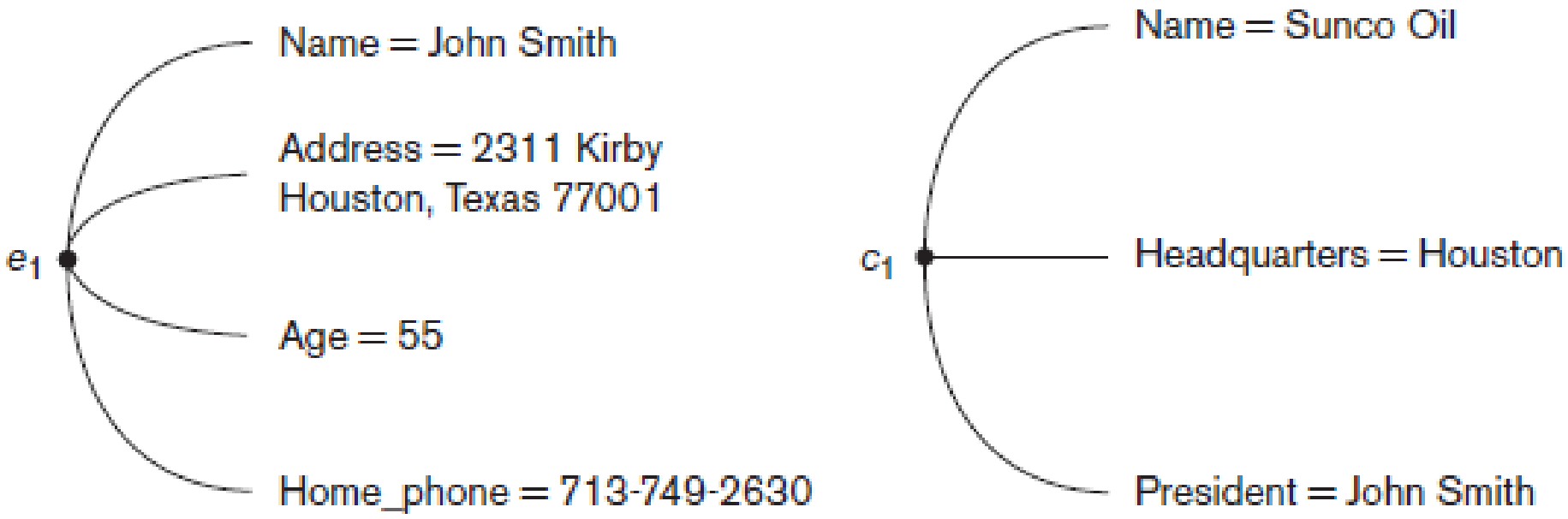


Figure 7.3
Two entities, EMPLOYEE e_1, and COMPANY c_1, and their attributes.

Entities, Attributes, Entity sets

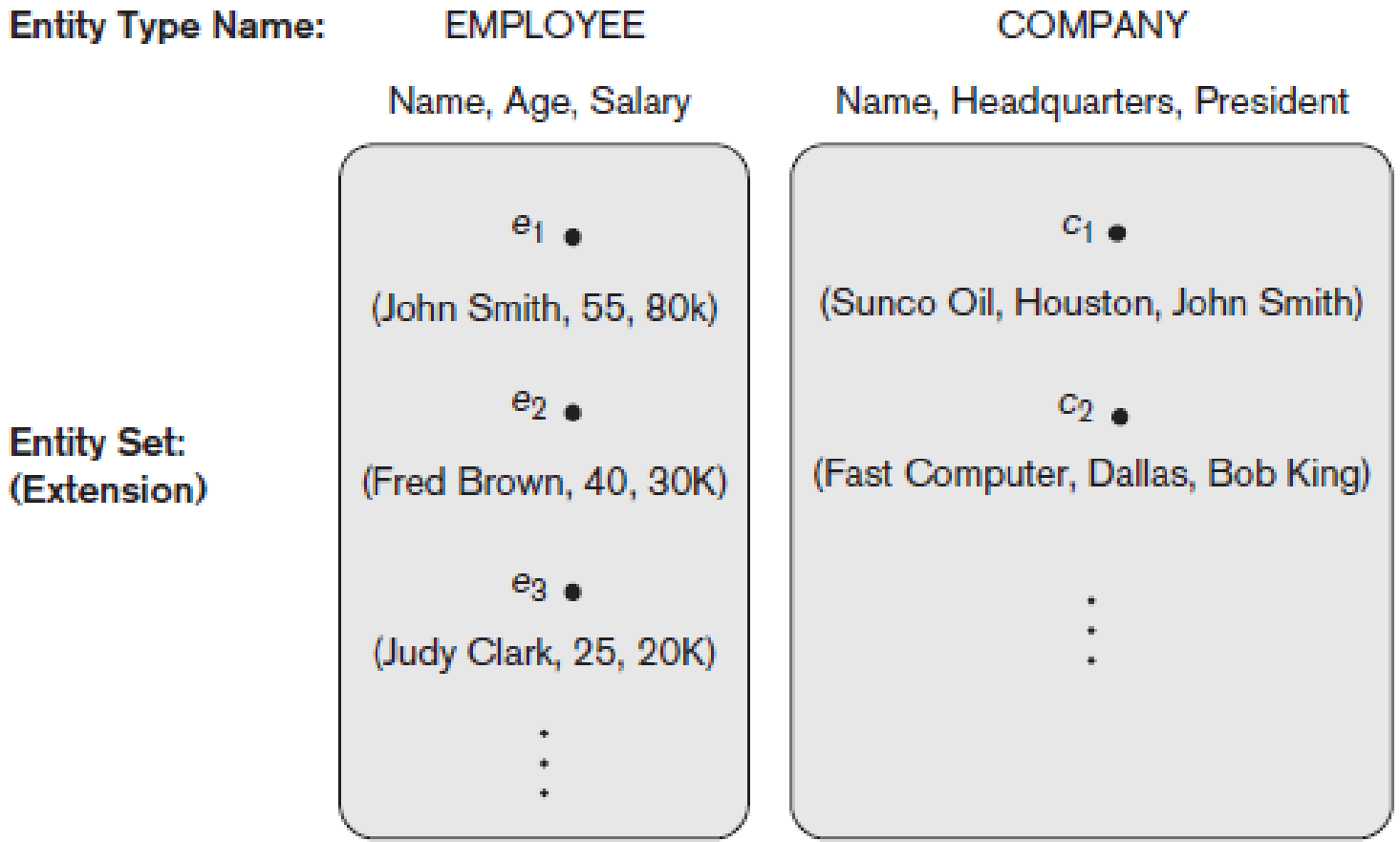
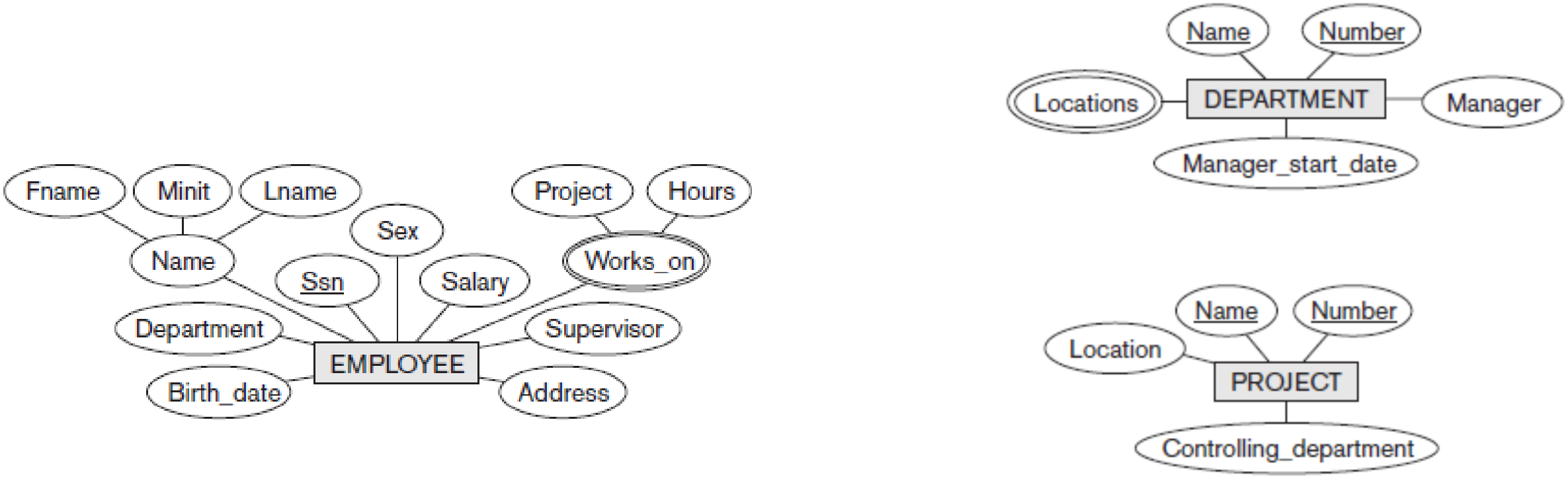


Figure 7.6
Two entity types, EMPLOYEE and COMPANY, and some member entities of each.

Key

- **Key** or **uniqueness constraint**
  - Attributes whose values are distinct for each individual entity in entity set
  - **Key attribute**
    - Uniqueness property must hold for every entity set of the entity type
- **Value sets** (or **domain of values**)
  - Specifies set of values that may be assigned to that attribute for each individual entity

Initial Conceptual Design of the COMPANY Database



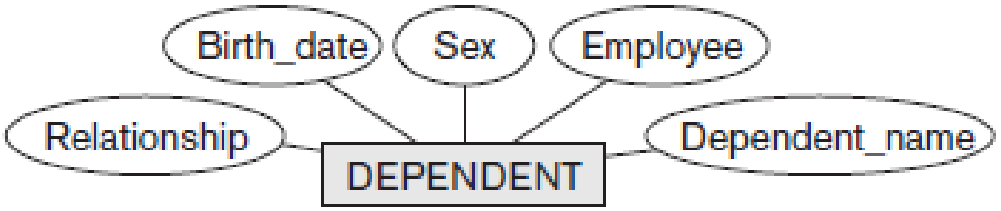


Figure 7.8
Preliminary design of entity types for the COMPANY database. Some of the shown attributes will be refined into relationships.

Relationship Types, Relationship Sets

- **Relationship**
  - When an attribute of one entity type refers to another entity type
  - Represent references as relationships not attributes
- **Degree** of a relationship type
  - Number of participating entity types
  - **Binary**, **ternary**
- Relationships as attributes
  - Think of a binary relationship type in terms of attributes

Relationship Types, Relationship Sets

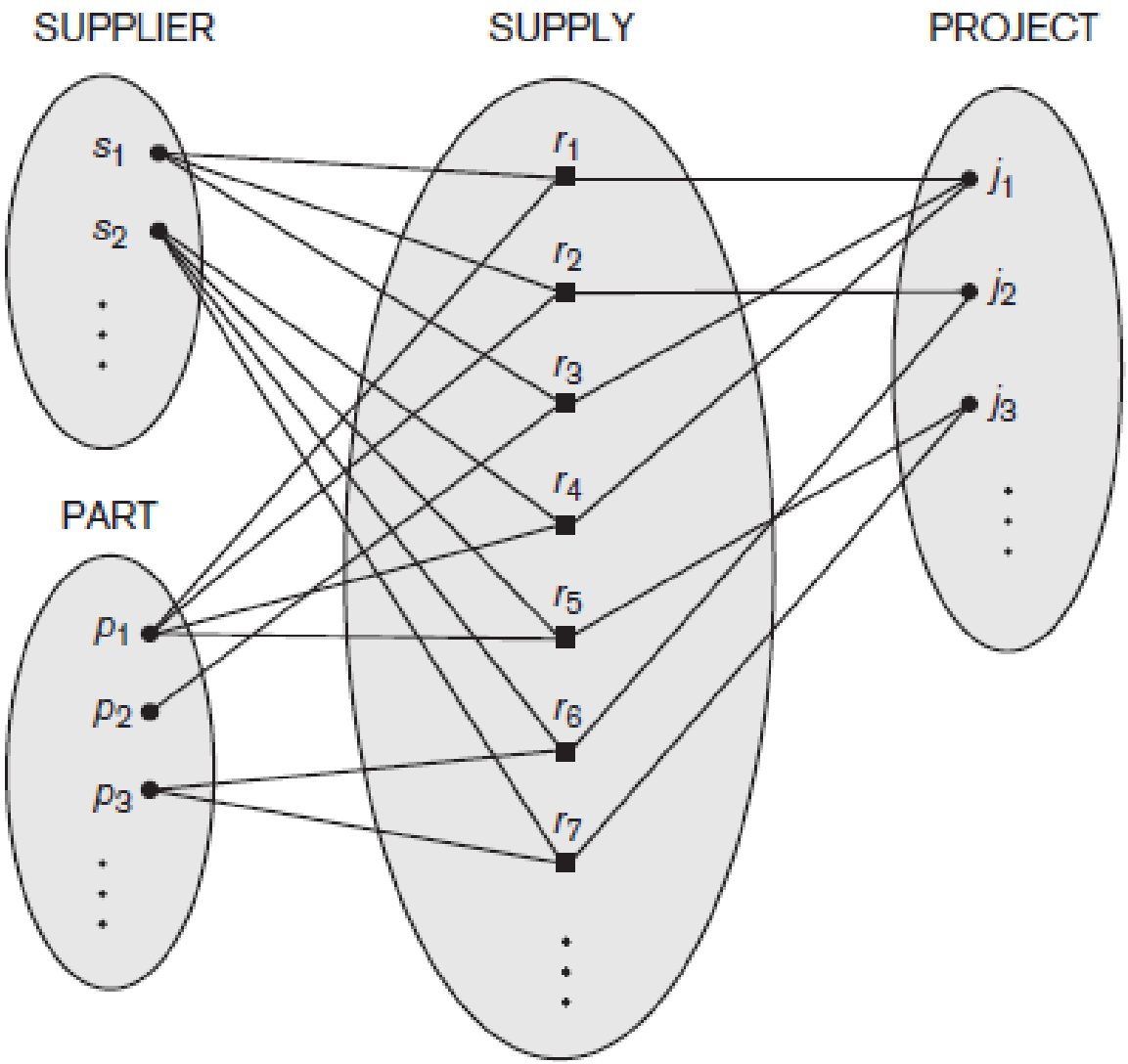
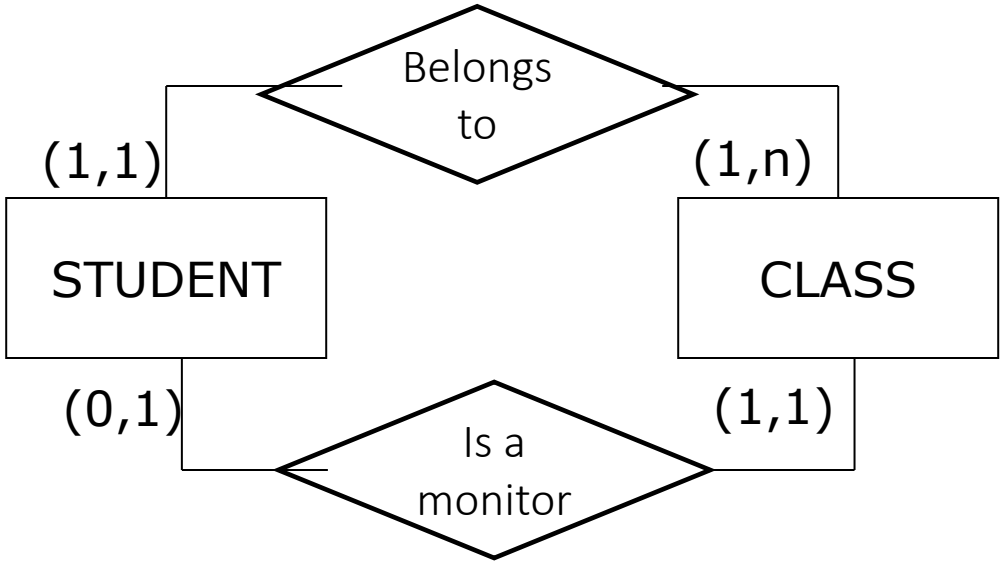


Figure 7.10
Some relationship instances in the SUPPLY ternary relationship set.

Relationship cardinality

- Shows the minimum and maximum entities associate with relationship.
- (min, max)



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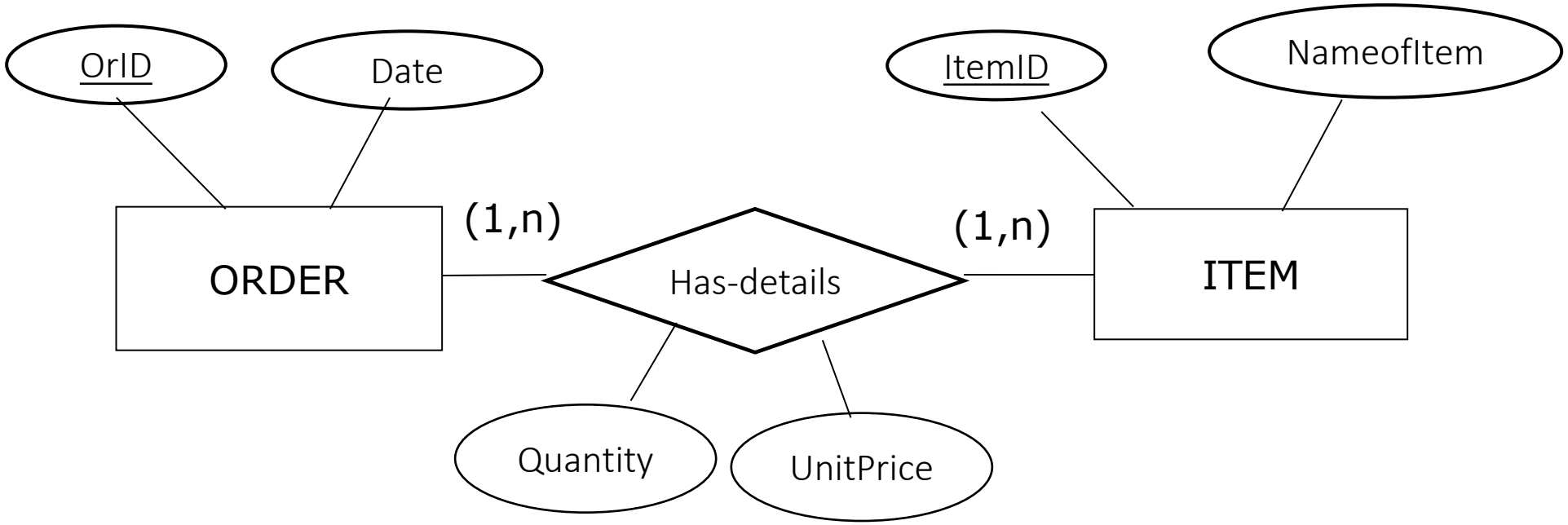

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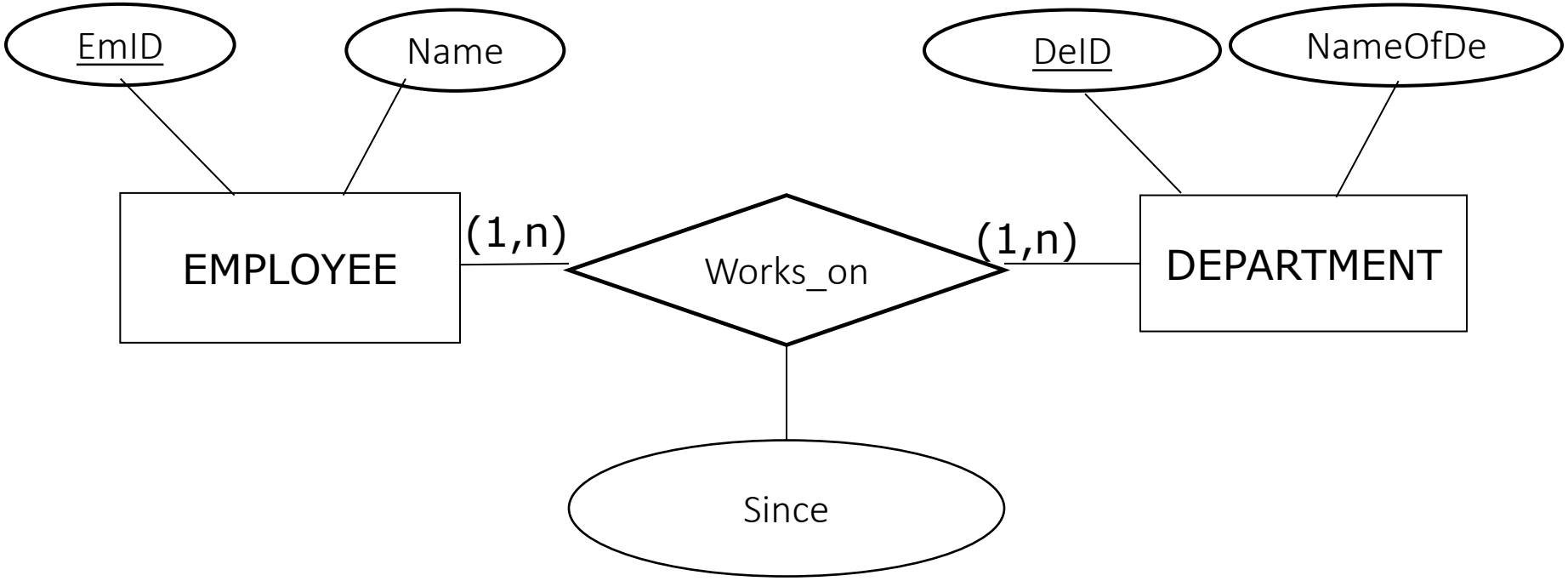
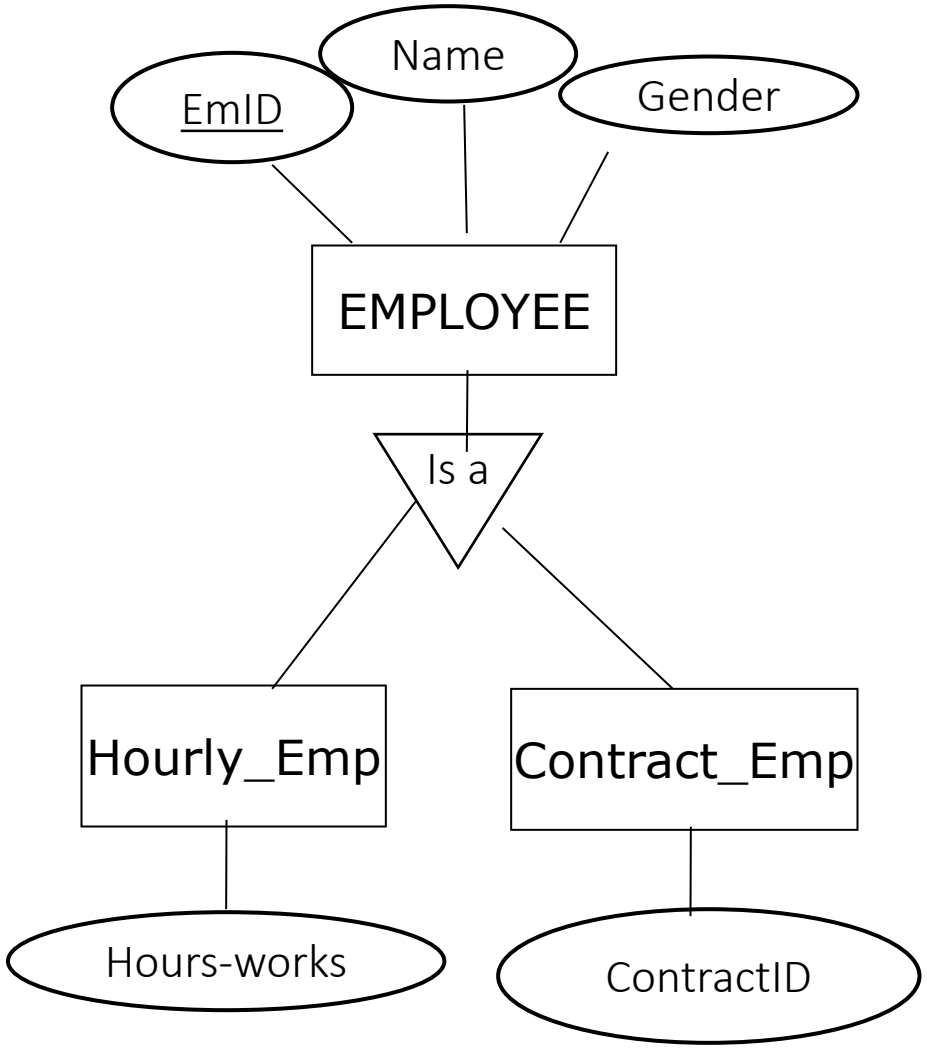


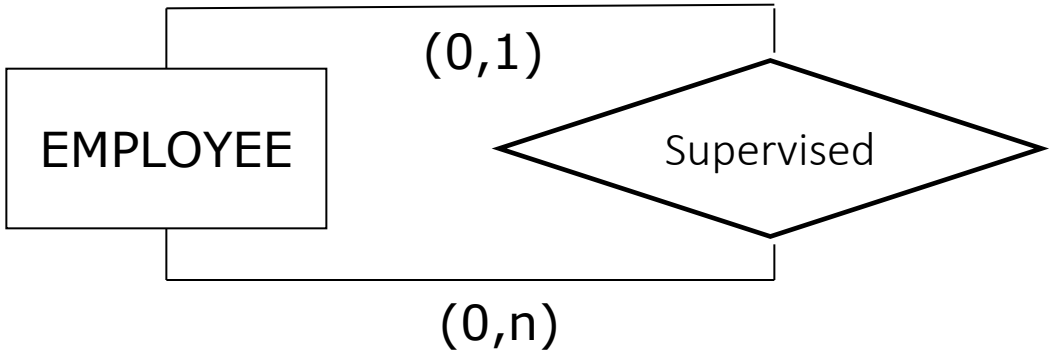
Extended Entity-relationship model

- Generalization and Specialization
- Recursive Relationship
- Weak entity
- Extended relationship

Generalization and Specialization

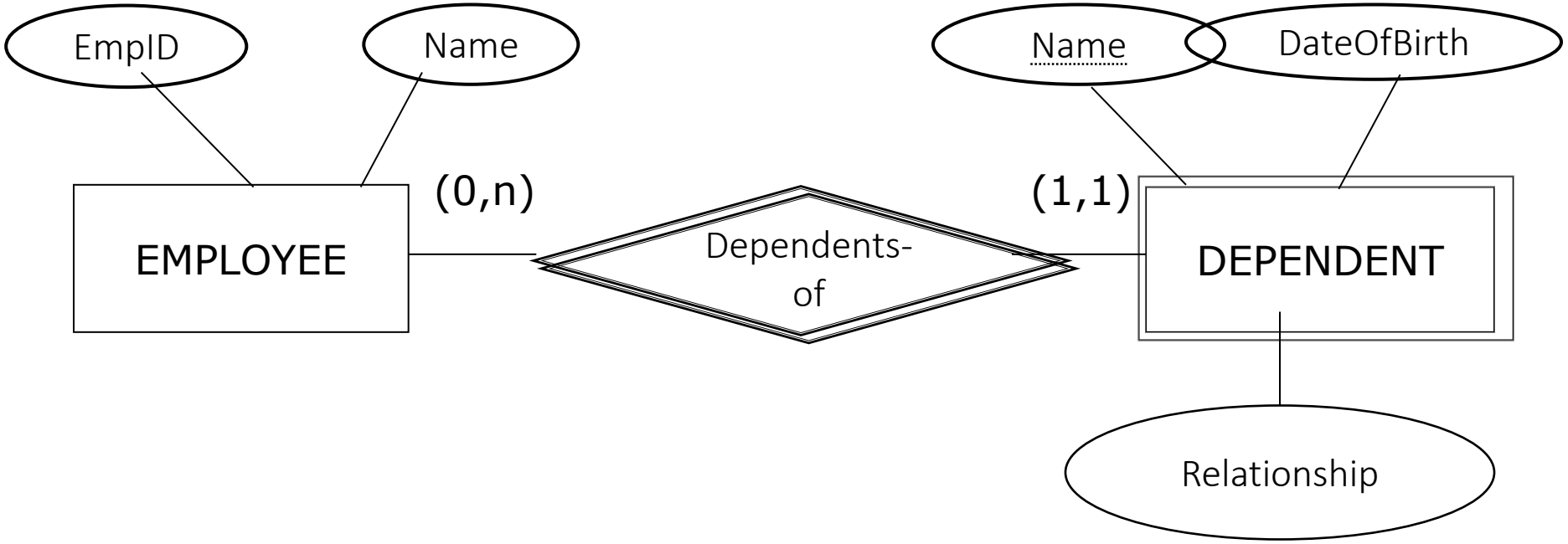


Recursive Relationship



Weak entity

- Entity has no key.
- It depends on other entity



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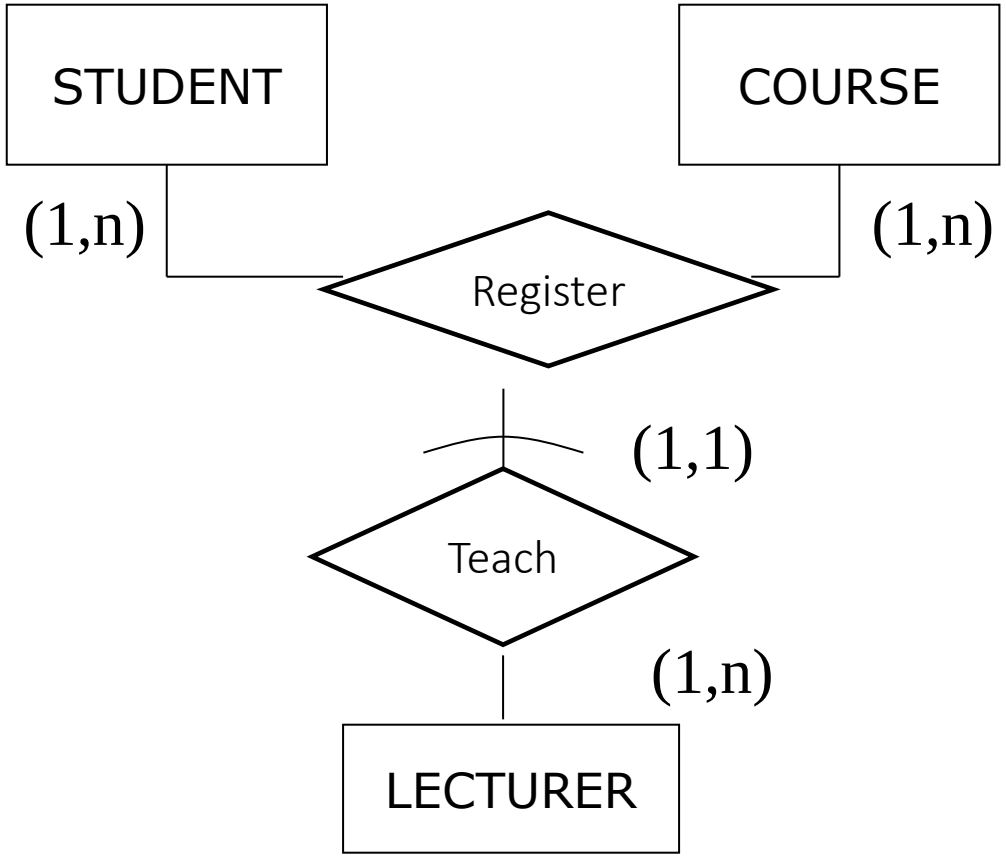
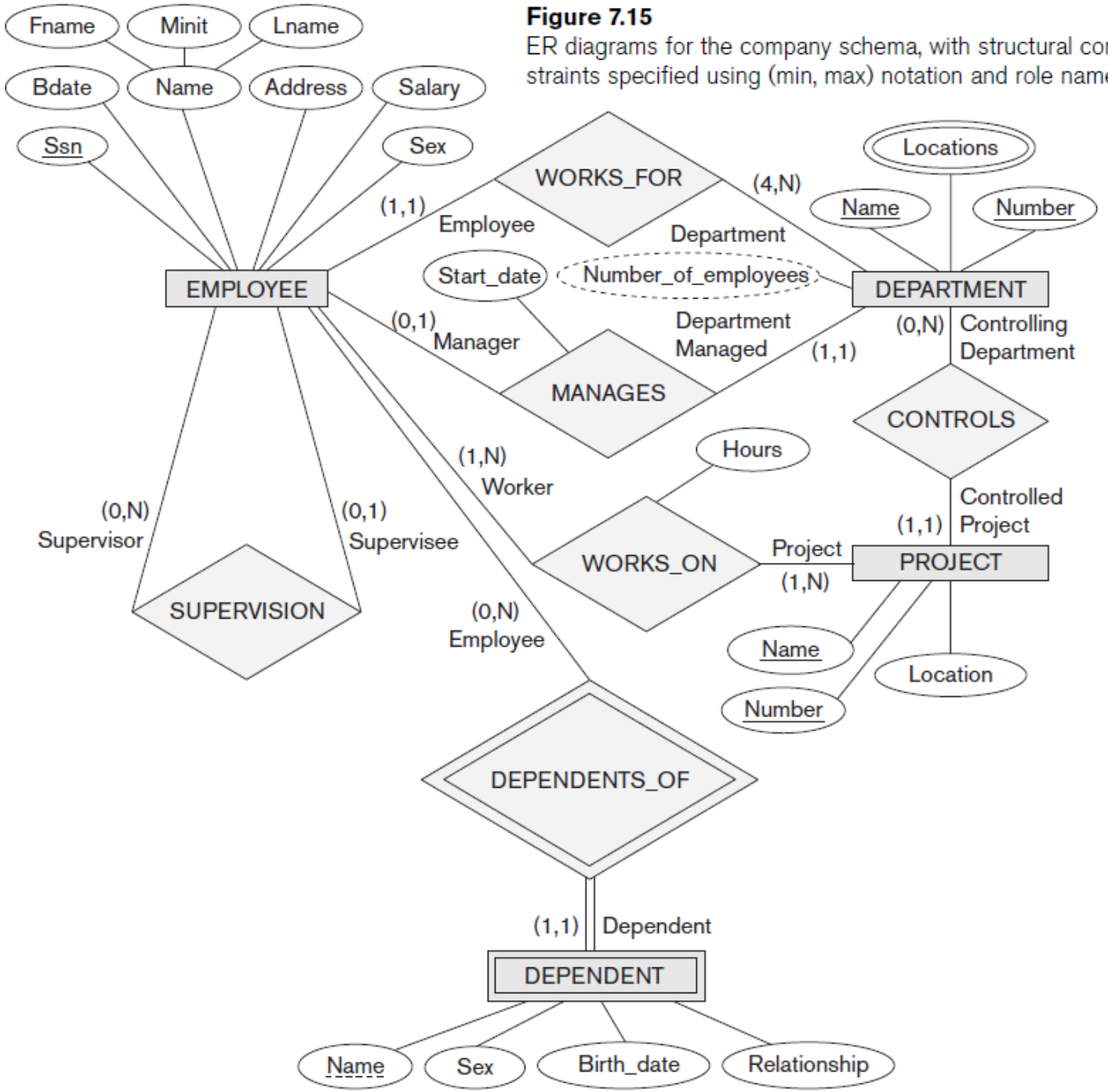


Figure 7.15
ER diagrams for the company schema, with structural constraints specified using (min, max) notation and role names.



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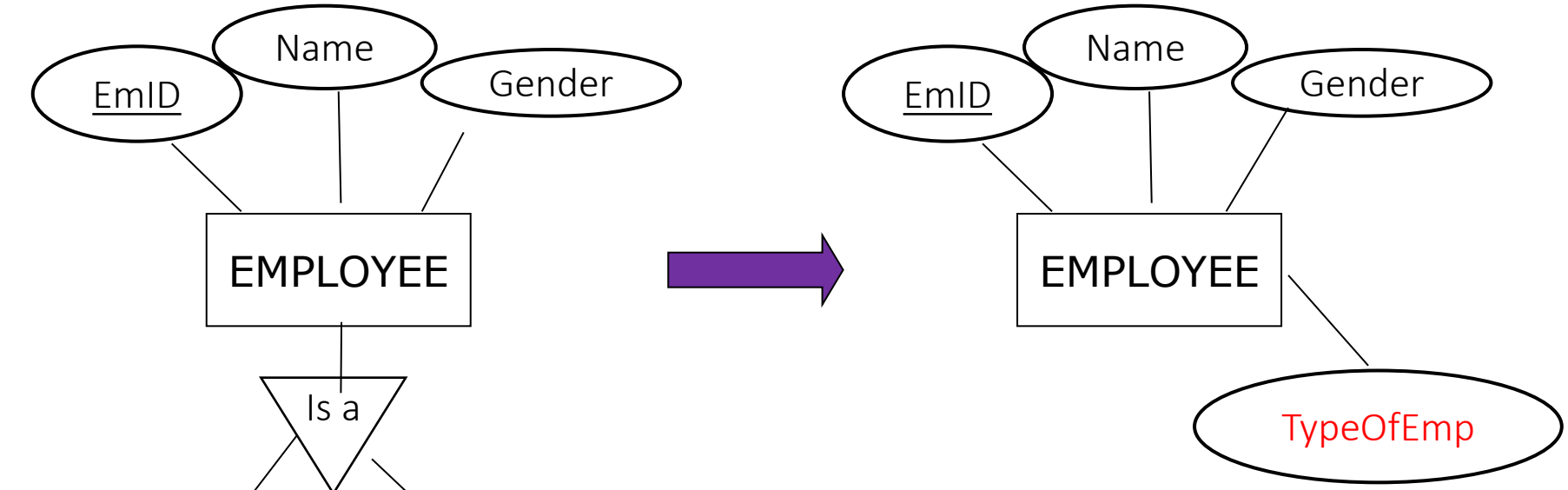
Converting ER model to Relational data model

1. Converting Class Hierarchies
2. Converting Entity set
3. Converting Relationships
4. Normalization

Converting Class Hierarchies

1. Specialization entities do not have their own attributes
   - Adding Type attribute to the Generalization entity
   - Adding constrains for the value of Type attribute



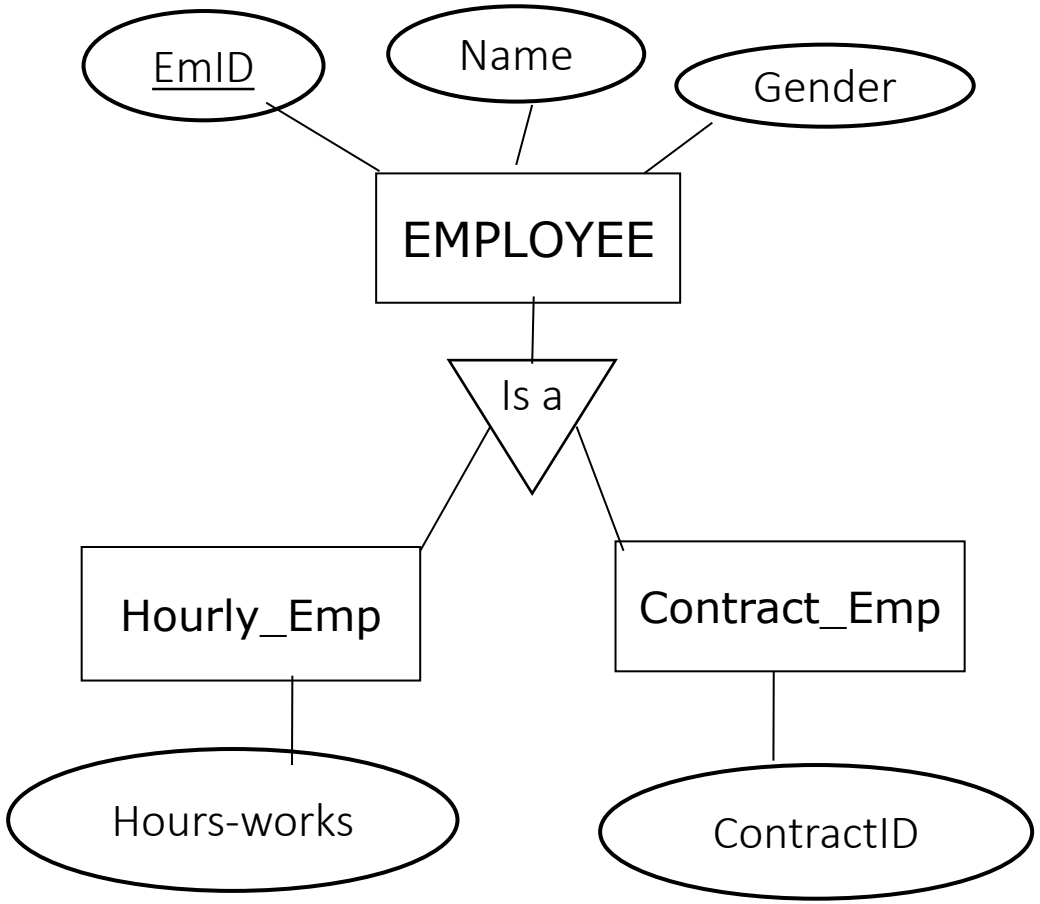
DOM(TypeOfEmp)={'Hourly_Emp', 'Contract_Emp'}

Hourly_Emp

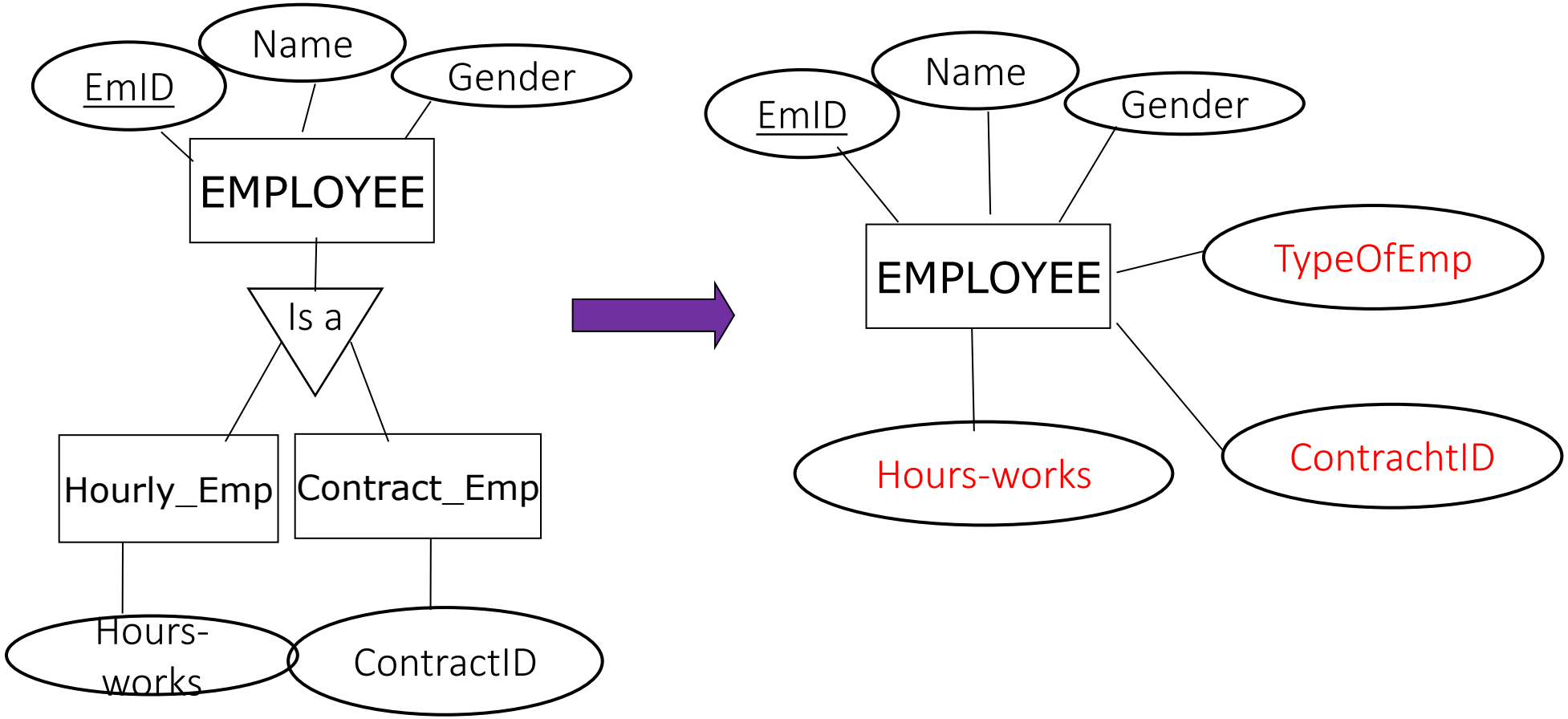
Contract_Emp

Converting Class Hierarchies

1. Specialization entities have a few attributes
   - Adding to the Generalization entity: Type attribute, specialization attributes
   - Adding constrains for: the value of Type attribute, specialization attributes



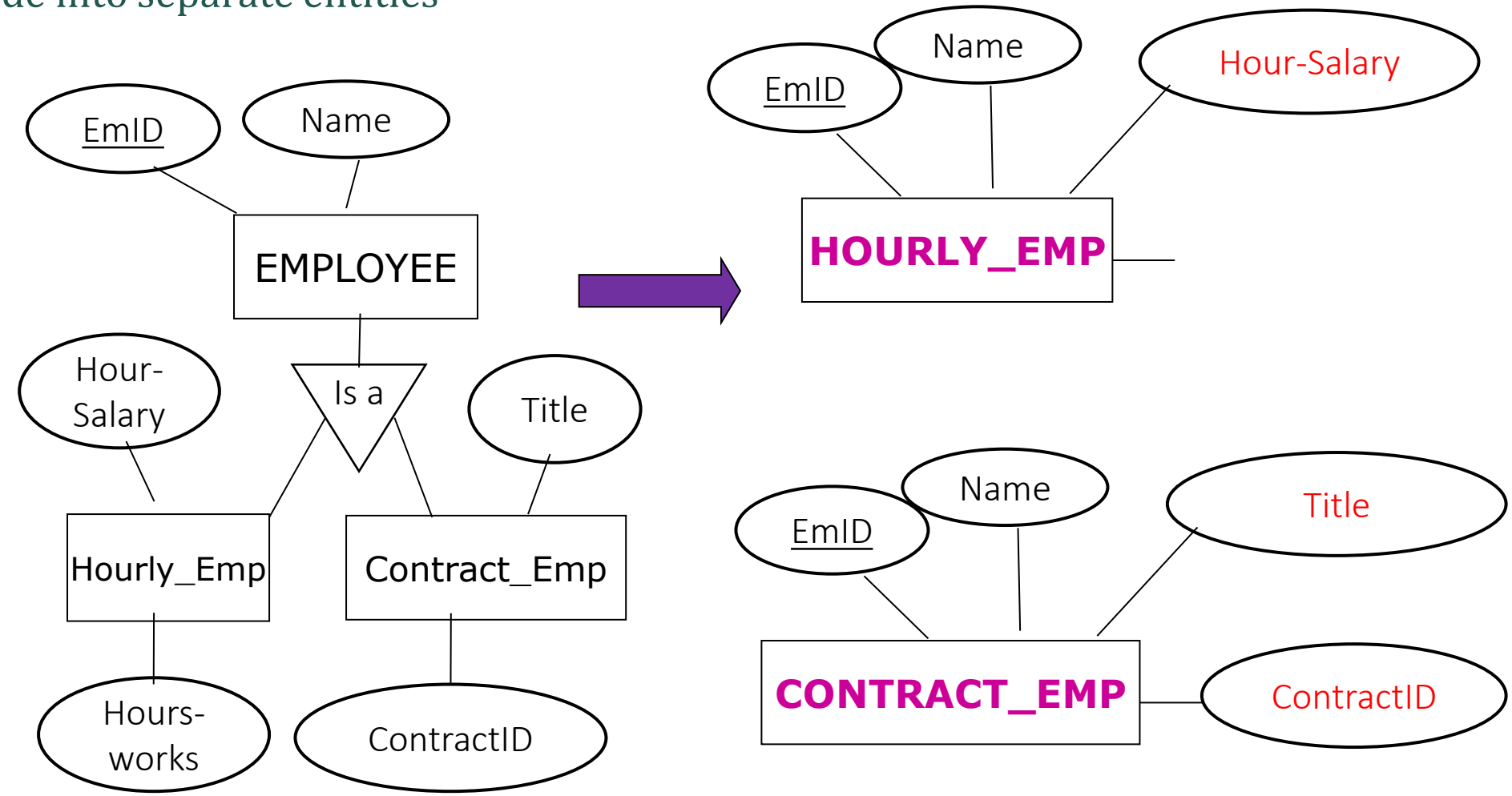
Converting Class Hierarchies



1. DOM(TypeOfEmp)={'Hourly_Emp', 'Contract_Emp'}
2. If TypeOfEmp=Hourly_Emp then Hours-works is enabled and ContractID is disabled
3. If TypeOfEmp=Contract_Emp then Hours-works is disabled and ContractID is enabled

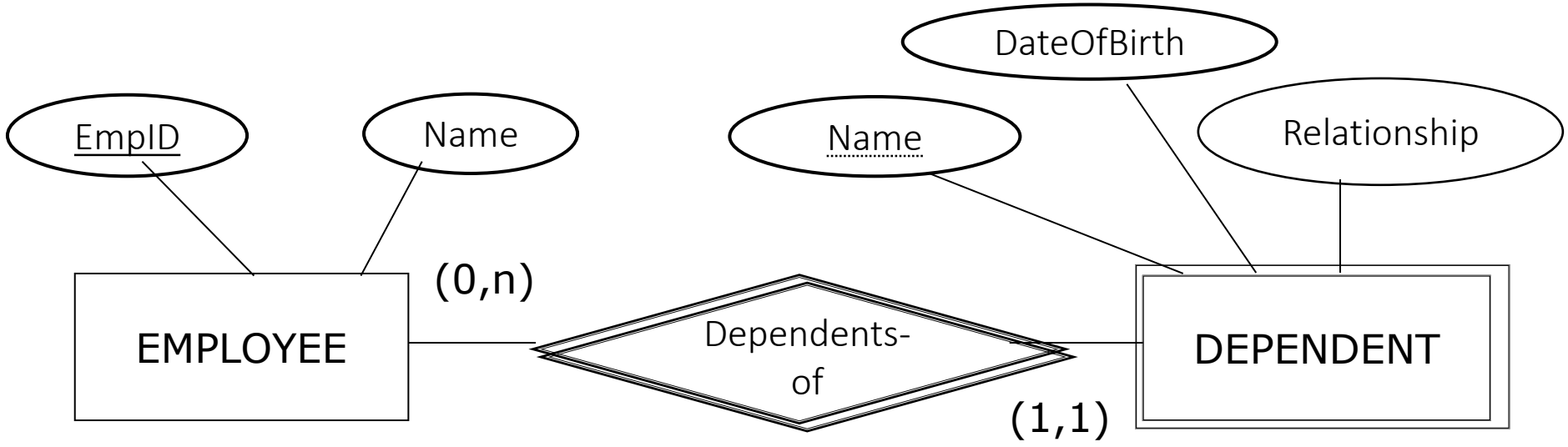
Converting Class Hierarchies

3. Specialization entities have a lot of attributes
   - Divide into separate entities



Converting Entity Set

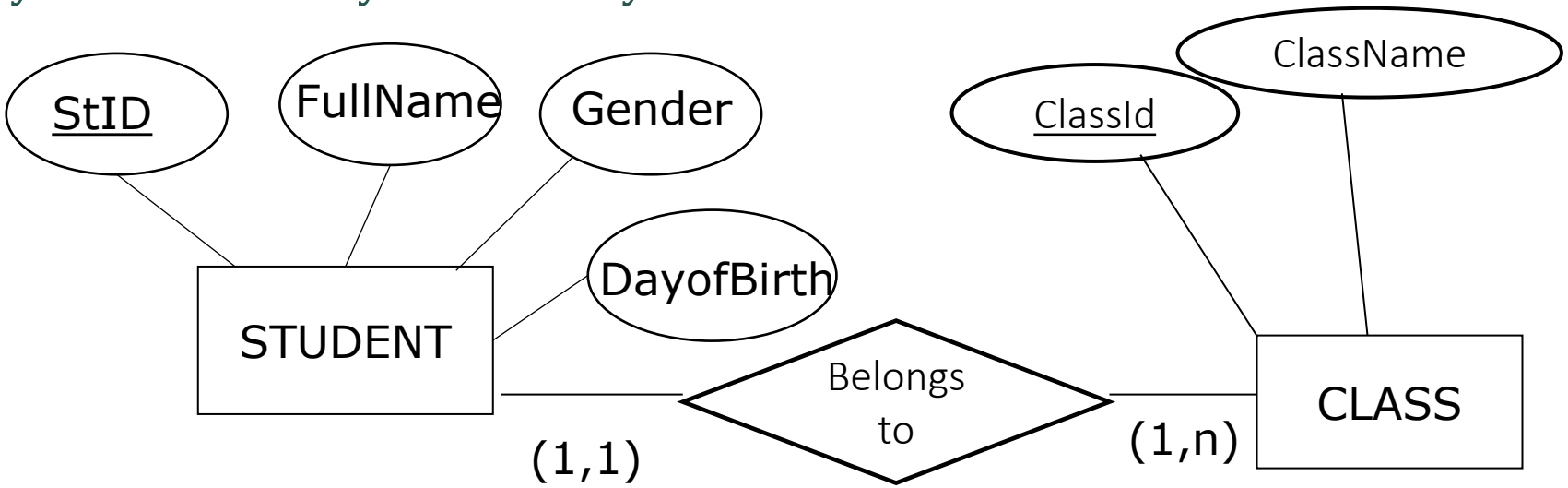
1. Weak Entity
   - Name of the table: Name of the entity
   - Atrributes of the table: Key of the strong related entity, and atrributes of the weak entity
   - Primary Key of the table: Key of the strong related entity and the difference attribute of weak entity



DEPENDENT(**<u>EmpID, Name,</u>** DateOfBirth, Relationship)

Converting Entity Set

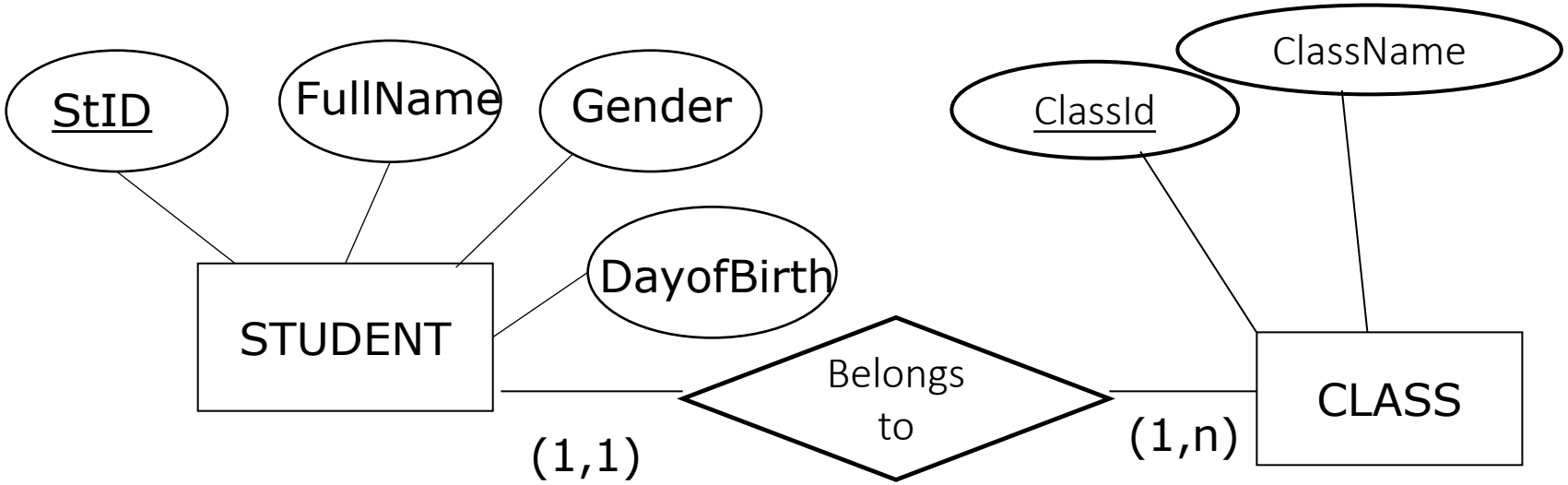
2. Entity set
   - Name of the table: Name of the entity
   - Atrributes of the table: Atrributes of the entity
   - Primary Key of the table: Key of the entity



1. STUDENT(**StID**, FullName, Gender, DayOfBirth)
2. CLASS (**ClassID**, ClassName)

Converting Relationship set

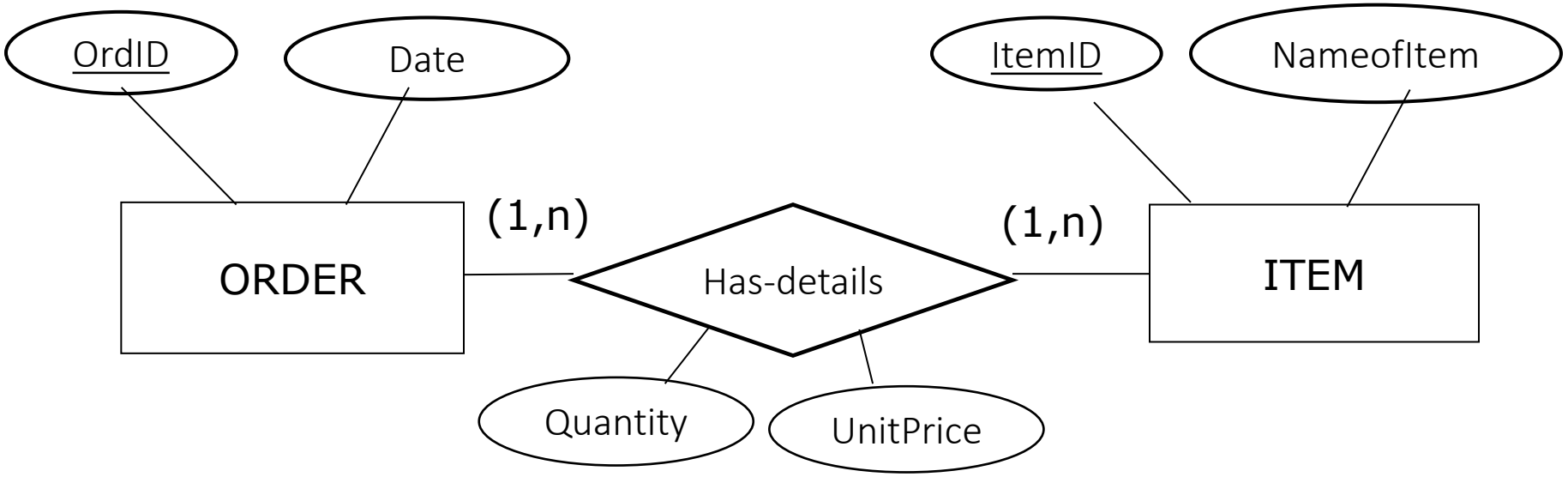
1. (1, 1) and (1,n) relatioship
   - Does not become a new table
   - Add key of the (1,n) entity to the table represented the (1,1) entity.



1. STUDENT(**StID**, FullName, Gender, DayOfBirth, **ClassID**)
2. CLASS (**ClassID**, ClassName)

Converting Relationship set

2. (1, n) and (1,n) relatioship
   - Does become a new table
   - Attributes of the new table: keys of related entities and attributes of the relationship
   - Primary key of the table: Key of related entities.



ORDER_ITEM (**OrdID, ItemID**, Quantity, UnitPrice)

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References

1. Ramez Elmasri, and Shamkant B. Navathe, *Fundamentals of Database Systems (7th Ed.)*, Addison-Wesley, 2016.

2. Nguyen Dang Ty, and Do Phuc, *Database*, Vietnam National University – HCMC, 2015.

3. George Tillmann, *Usage-Driven Database Design,* Apress, 2017.

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